

Attempts to Mitigate Algorithmic Collusion in LLMs

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Abstract

FILL ONCE ALL RESULTS ARE READY.

1 Algorithmic Collusion in LLMs

Fish, Gonczarowski, and Shorrer study whether Large Language Model (LLM)-based pricing agents can autonomously generate collusive-like outcomes in repeated market environments. Their central object of study is *autonomous algorithmic collusion*: supracompetitive outcomes produced by learning or AI-based algorithms without explicit human design to collude. This framing builds on a growing literature on algorithmic pricing and competition policy, where regulators and researchers have raised concerns that automated pricing systems may sustain high prices even without direct communication among firms [Harrington, 2018, Ezrachi and Stucke, 2020, Calvano et al., 2020].

The paper situates itself against earlier work on algorithmic collusion, especially studies of Q-learning pricing agents. Calvano et al. [2020] show that Q-learning algorithms can learn supracompetitive pricing in repeated Bertrand environments. Fish et al. emphasize, however, that these earlier algorithmic agents face practical barriers where they often require long training periods, and may be exploitable by competitors. This motivates the paper’s key policy question which can be summed as: do modern AI systems that avoid these barriers also exhibit autonomous algorithmic collusion?

The authors argue that LLMs are an especially important test case. Unlike classical reinforcement-learning pricing algorithms, LLMs are pretrained on broad datasets and can be deployed without costly environment-specific training. They can also operate in many different environments using natural-language instructions. These features make them plausible candidates for real-world pricing agents. At the same time, however, they are opaque, stochastic, and difficult to interpret, which creates both scientific and regulatory challenges. The paper therefore treats LLMs not merely as tools for simulating humans, but as a new kind of economic agent whose behavior must be experimentally analyzed.

Fish et al. explicitly ask three research questions: whether LLM-based pricing agents behave in a collusive-like manner and reach supracompetitive prices; whether they do so without being instructed or intended to collude; and, if so, what mechanisms drive this behavior. They claim to be the first to experimentally study autonomous algorithmic collusion by LLM-based pricing agents.

1.1 Experimental Design

The main experiment uses a repeated Bertrand oligopoly environment following Calvano et al. [2020]. Each firm delegates pricing to an independent LLM-based agent. In every period, each agent sets a price, demand is realized, and the agent observes the prices chosen by all firms as well as its own quantity sold and profit. Agents do not observe the underlying demand function and cannot communicate directly or indirectly except through the prices they set.

The environment uses a logit demand model. If firms $1, \dots, n$ set prices $p_1, \dots, p_n$, firm i ’s demand is

$$q_i = \beta \frac{\exp((a_i - p_i/\alpha)/\mu)}{\sum_{j=1}^n \exp((a_j - p_j/\alpha)/\mu) + \exp(a_0/\mu)}.$$

Profits are given by

$$\pi_i = (p_i - \alpha c_i) q_i.$$

In the main experiments, the authors use symmetric firms with $a_i = 2$, outside option $a_0 = 0$, horizontal differentiation parameter $\mu = 0.25$, and marginal cost $c_i = 1$, following Calvano et al. [2020]. They set $\beta = 100$ so that quantities appear in a natural numerical scale to the LLM. They also randomize the currency-scaling parameter $\alpha \in \{1, 3.2, 10\}$ because LLMs may not be invariant to changes in units.

A single experimental run consists of 300 periods and the LLM prompt contains five main components: a prompt prefix stating the high-level goal, basic market information including marginal cost and a price ceiling, market history from the previous 100 periods, the agent’s own “plans and insights” from the previous period, and output instructions requiring the agent to generate reasoning, update its plans, and set a price. This design gives the otherwise memoryless LLM some continuity of thought across periods.

The authors compare two prompt prefixes, denoted P1 and P2. Both instruct the agent to maximize long-run profit, but they differ in their final phrasing. P1 emphasizes avoiding actions that undermine profitability, while P2 explicitly mentions exploration, risky or aggressive pricing, and the fact that pricing lower than a competitor can increase quantity sold. The exact prompt summaries are given in Appendix A. This prompt manipulation is important because one of the paper’s claims is that seemingly innocuous language can substantially alter market outcomes.

Before running the main duopoly experiment, the authors evaluate candidate LLMs in a monopoly setting. They view strong monopoly performance as a necessary condition for meaningful multi-firm pricing experiments. Based on this preliminary evaluation, they select GPT-4 as the primary model for the main duopoly experiment.

For each prompt prefix, P1 and P2, the authors conduct 21 independent 300-period duopoly runs. They measure pricing and profits primarily over the final 50 periods, comparing outcomes to the Bertrand–Nash price p^{Nash} of the one-shot game and to the monopoly price p^M that would maximize total profit if both firms were controlled by a single monopolist. Refer to Appendix B for more details on how we compute those in our specific setup.

1.2 Findings

The central finding is that LLM-based pricing agents quickly and consistently reach supracompetitive prices and profits. Both P1 and P2 lead to prices above the Bertrand–Nash benchmark. However, P1 produces substantially higher prices than P2, sometimes even above the monopoly price. The profit results follow the same pattern also, where both prompts generate supracompetitive profits, but P1 produces total profits closer to monopoly levels. This is interesting because neither prompt instructs the agents to collude, retaliate, avoid competition, or coordinate with competitors.

The contrast between P1 and P2 is one of the paper’s most important empirical observations. P2 includes language that makes undercutting more salient by telling the agent that lower prices can increase quantity sold. P1 instead reinforces long-run profitability. The authors argue that the difference is not simply that P2 gives the agent new economic knowledge and they show that P1 agents already understand that lower prices can increase quantity sold. Rather, the prompt wording appears to change the strategic frame through which agents interpret the environment.

The authors also report robustness results in appendices. Their main findings persist under noise, asymmetric demand, heterogeneous pricing agents, instructions to discount the future, and newer-generation LLMs. They also extend the analysis to first-price auctions, another domain where prior work found autonomous collusion among Q-learning agents.

1.3 Reward-Punishment, and Textual Reasoning.

After documenting supracompetitive pricing, Fish et al. ask what mechanisms sustain it. They focus on reward-punishment strategies, a classic explanation for tacit collusion in repeated oligopoly games. In such strategies, agents maintain high prices when rivals cooperate but punish deviations, such as price cuts, with lower prices for some time. This logic appears in the theoretical literature on collusion [Stigler, 1964, Friedman, 1971, Green and Porter, 1984, Harrington, 2018] and in prior work on Q-learning pricing agents [Calvano et al., 2020].

The authors conduct two complementary analyses tailored for the LLM case. **First**, an off-path analysis which uses the fact that LLM agents produce textual reasoning and plans. The authors search these generated plans for price-war-related reasoning. They find that agents, especially under P1, more often express concern about avoiding price wars. However, textual correlation alone does not establish that such reasoning causally affects pricing.

To address this, the authors introduce an “implantation” method. They reset completed simulations to earlier states and replace an agent’s plans and insights with text expressing concern about avoiding a price war. They then rerun the experiment from that point and compare the resulting price to the original price. Prices after implantation are significantly higher, which suggests that price-war-concerned reasoning **causally** contributes to higher pricing. This method is one of the paper’s methodological contributions: because AI agents can be reset

and perturbed, researchers can perform counterfactual behavioral experiments that are impossible with human subjects.

Secondly, an on-path analysis that uses regression to examine whether observed prices follow a reward-punishment pattern. The authors estimate a model of the form

$$p_{i,r}^t = \alpha_{i,r} + \gamma p_{i,r}^{t-1} + \delta p_{-i,r}^{t-1} + \varepsilon_{i,r}^t \quad (1)$$

where $p_{i,r}^t$ is agent i 's price in period t of run r , $p_{i,r}^{t-1}$ is its own previous price, $p_{-i,r}^{t-1}$ is its competitor's previous price, and $\alpha_{i,r}$ is a firm-run fixed effect. A positive coefficient on the competitor's previous price indicates that the agent raises prices after the competitor raises prices and lowers prices after the competitor lowers prices. A positive coefficient on the agent's own previous price indicates persistence, meaning rewards and punishments decay gradually rather than disappearing immediately.

The regression results are consistent with reward-punishment behavior. Agents respond positively to competitors' previous prices, and their own prices are sticky over time. These effects are stronger under P1, the prompt that also produces higher prices, higher profits, and greater price-war concern in the textual analysis. Taken together, the off-path and on-path evidence suggests that LLM-based pricing agents avoid price reductions partly because they anticipate or "fear" retaliation, and when lower prices occur, their realized behavior resembles punishment dynamics.

Finally, the authors broaden the textual analysis using clustering methods. They find that prompt wording systematically changes the types of reasoning agents produce. P1 agents more often emphasize sustaining price levels and reacting to competitors, while P2 agents more often emphasize undercutting and exploration. Using the implantation method again, the authors link undercutting-oriented reasoning to lower prices. Thus, the paper argues not only that LLM agents produce supracompetitive outcomes, but also that their natural-language reasoning contains strategically meaningful content that can help explain their behavior.

Overall, Fish et al. provide evidence that LLM-based pricing agents can autonomously generate collusive-like outcomes in repeated oligopoly environments. Their contribution is not only empirical but also methodological. Empirically, they show that LLM agents reach supracompetitive prices and profits without being instructed to collude, and that small prompt differences can substantially change the degree of supracompetitive pricing. They also show how LLM-generated reasoning can be analyzed and causally perturbed through implantation experiments. The paper therefore raises a regulatory concern: if natural-language prompts can meaningfully affect market conduct, then regulating LLM-based pricing agents may require attention not only to outcomes and code, but also to instructions, reasoning traces, and behavioral responses across market histories.

2 Project Overview

We take Fish et al.'s regulatory concern as the starting point for our project. If LLM-based pricing agents can produce supracompetitive outcomes through natural-language instructions and market histories, then a natural next question is:

Can regulatory interventions reduce prompt-induced supracompetitive pricing by LLM pricing agents, and which intervention channels perform best?

Thus, our project has two goals. First, we attempt to reproduce the qualitative P1/P2 effect reported by Fish et al. using a lower-cost model (since GPT-4 is extremely expensive). Second, after establishing a baseline, we introduce several regulatory interventions and evaluate whether they reduce supracompetitive pricing.

The interventions we study target different regulatory channels. Some modify the agents' instructions, some modify realized payoffs through fines, and some introduce competitive price signals or an economically active additional competitor. We then analyze these interventions using the same basic outcome measures as Fish et al. The rest of the report is organized as follows. In section 3, we situate the project within the literatures on algorithmic collusion, LLM agents, and regulation of pricing algorithms. In Section 4, we describe our model-selection constraints and try to reproduce the original P1/P2 experiment and results. In section 5, we define the regulatory interventions we test. Finally, in Section 6, we report the wrap-up the results, the effects of the interventions, and trajectory-level analyses inspired by Fish et al.'s methodology.

3 Related Work

Our work builds on four lines of research.

Algorithmic pricing and LLM collusion. As we already established in prior sections, Calvano et al. [2020] provide the canonical simulation evidence that Q-learning pricing agents can sustain supracompetitive prices in repeated oligopoly environments without explicit communication. Fish et al. [2025] extend this concern to LLM-based pricing agents, using an economic environment that closely follows Calvano et al. but replacing task-specific reinforcement learning with pretrained, prompt-driven agents that use natural-language plans and memory.

Reward-punishment mechanisms. A central interpretation in the collusion literature is that supracompetitive prices can be sustained by reward-punishment strategies in repeated interaction (Stigler [1964], Friedman [1971], Green and Porter [1984], Harrington [2018]). In a repeated pricing game, firms may avoid short-run profitable price cuts because such deviations can trigger future punishment, such as a price war. Calvano et al. [2020] interpret the behavior of Q-learning pricing agents through this lens, and Fish et al. [2025] explicitly ask whether LLM agents reason about price-war risks and respond to competitors’ past prices in ways consistent with reward-punishment behavior. This will also be one of the scopes of the project as we will apply similar analyses.

Antitrust enforcement and expected damages. In addition to our own heuristic-based regulatory schemes, we adopt a regulatory intervention that is motivated by Harrington’s dynamic model of cartel pricing under antitrust enforcement Harrington [2005]. Harrington studies a cartel that chooses a price path while facing probabilistic detection and penalties. The key insight for our purposes is the distinction between fixed fines and price-sensitive damages. Fixed fines do not scale with the size or persistence of the overcharge, whereas damages accumulate with the harm caused by supracompetitive prices. In Harrington’s model, fixed fines alone do not affect the steady-state cartel price, while the steady-state cartel price is decreasing in the damage multiple and the probability of detection.

4 Reproducing Fish et al.’s Results

A first step in our project was to reproduce the baseline phenomenon reported by Fish et al. [2025]. This required choosing a model that could plausibly behave like the LLM pricing agents in their original experiment. Fish et al. select GPT-4 after first evaluating candidate models in a single-agent monopoly environment: their logic is that an LLM that cannot learn reasonable pricing behavior when facing no competitor is unlikely to provide a meaningful test of strategic pricing behavior in a duopoly environment.

Directly reproducing their main experiments with GPT-4 through the API was not feasible for us because of cost. We therefore tested several modern LLMs as candidate substitutes, including Gemini-3-27B, GPT-4o, GPT-4o-mini, DeepSeek-V3, and DeepSeek-V3.1. Among the models we tried, DeepSeek-V3 failed the preliminary monopoly test, while DeepSeek-V3.1 produced usable single-agent behavior. Due to cost constraints, we therefore fixed DeepSeek-V3.1 as the model for the rest of our experiments. Even with this cheaper model, the full experimental pipeline cost roughly \$100 through the DeepInfra API. Since we did not have sufficient local compute to run a comparable model ourselves, this API-based setup was the most practical compromise between fidelity to Fish et al.’s design and available resources.

4.1 Experimental Design

For the reproduction experiment, we follow Fish et al.’s experimental design 1.1. We use the same repeated Bertrand duopoly environment, the same prompt structure, and the same agent architecture. In particular, we do not modify the prompts, the market-history format, the plans-and-insights mechanism, or the interaction protocol between agents. Each agent independently receives its own prompt, chooses a price, observes the resulting market history, and carries forward its own plans and insights to the next period. As in Fish et al., agents cannot communicate directly with each other; their only strategic interaction is through the prices they set and the market outcomes that follow.

This reproduction step is important because our regulatory experiments build on the baseline finding that prompt-driven LLM pricing agents can reach supracompetitive prices. Before testing interventions, we therefore first verify whether the same qualitative pattern appears under our model and API constraints.

4.2 No-Regulator Reproduction Results

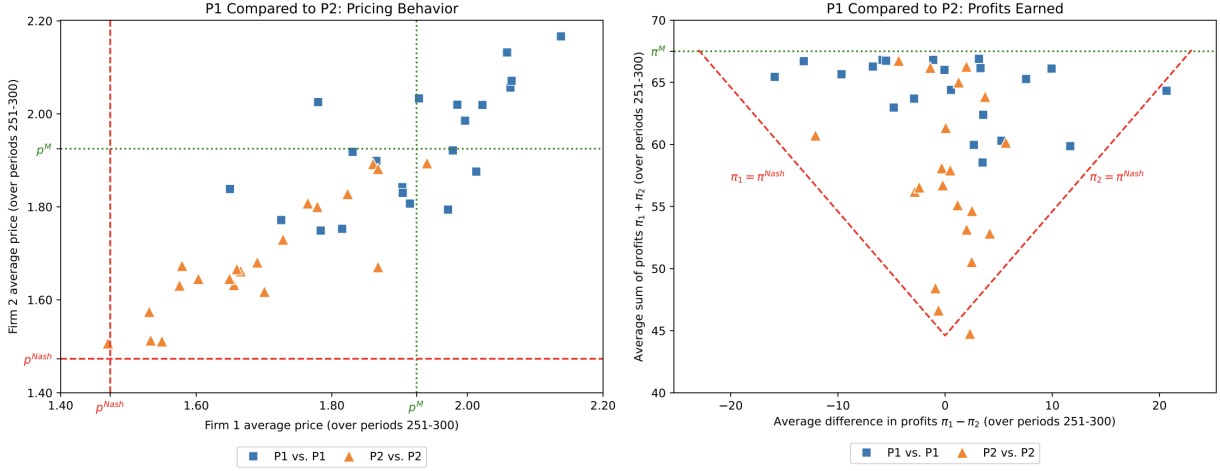


Figure 1: Original duopoly results from . Fish et al. [2025], showing that both P1 and P2 produce supracompetitive outcomes, with P1 reaching higher prices and profits closer to monopoly levels.

Figure 1 shows the original main duopoly result from Fish et al. [2025], while Figure 2 reports our reproduction using DeepSeek-V3.1. In both figures, the left panel plots the average normalized price of firm 1 against the average normalized price of firm 2 over the final 50 periods. The red dashed lines mark the one-shot Bertrand–Nash benchmark, and the green dotted lines mark the joint-profit-maximizing monopoly benchmark. The right panel presents the corresponding profit outcomes: the horizontal axis measures the average profit difference, $\pi_1 - \pi_2$, while the vertical axis measures total average profit, $\pi_1 + \pi_2$. Fish et al.’s original GPT-4 results show strong(er) supracompetitive pricing under both prompt conditions, with P1 producing substantially higher prices and profits than P2. Many P1 runs are close to monopoly-level profits, and some prices even exceed the monopoly benchmark. Hence, in their work the outcomes are *collusive-like* or *supracompetitive*.

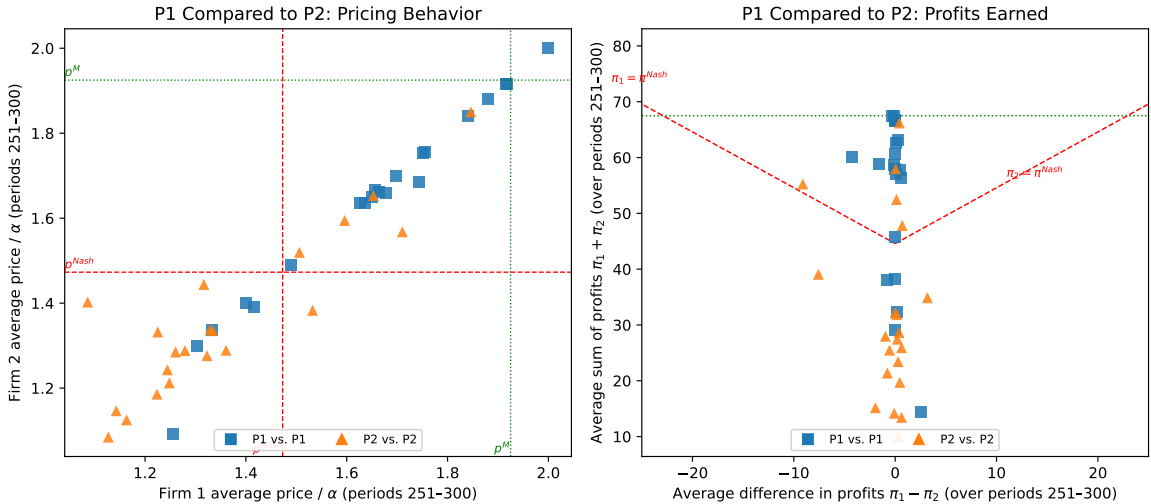


Figure 2: Our no-regulator DeepSeek-V3.1 reproduction, showing a statistically significant P1-P2 separation but weaker monopoly-level pricing than in Fish et al.’s original GPT-4 results.

Our DeepSeek-V3.1 reproduction recovers the main direction of Fish et al.’s prompt effect, but at a weaker level. P1 still produces significantly higher prices and profits than P2, as summarized in Table 1. However, the absolute pricing level is lower than in Fish et al.’s original GPT-4 experiment. In our runs, the normalized Nash and

Test	Fish et al. GPT-4	Our DeepSeek-V3.1
Avg. price, P1 vs. P2	$p < 0.00001$	$p = 0.00006$
Total profit, P1 vs. P2	$p < 0.05$	$p = 0.00005$

Table 1: Comparison between the statistical tests reported by Fish et al. [2025] and our no-regulator DeepSeek-V3.1 reproduction, using two-sided Welch’s t -tests over the final 50 periods.

monopoly benchmarks are $p^{Nash}/\alpha = 1.473$ and $p^M/\alpha = 1.925$. Under P1, the mean normalized price is above Nash, and 76% of P1 runs exceed the Nash benchmark, but only $\approx 5\%$ exceed the monopoly benchmark. Under P2, the mean normalized price is below Nash, only 29% of runs exceed Nash, and no runs exceed the monopoly benchmark.

The geometry of Figure 2 reveals an additional difference not observed in Fish et al.’s work. Our price points lie close to the diagonal. Meaning, when firm 1 sets a high average price, firm 2 usually does as well, and when one firm sets a lower price, the other tends to follow. This suggests stronger cross-firm price tracking or responsiveness. The agents often move together, but many runs remain closer to Nash than to monopoly. Thus, DeepSeek-V3.1 exhibits coordinated and prompt-sensitive pricing behavior, but weaker supracompetitive extraction than the GPT-4 agents in Fish et al.

Overall, the no-regulator reproduction provides a usable baseline for our regulatory experiments. It confirms that the qualitative prompt effect survives under our cheaper model and API constraints: P1 generates significantly higher prices and profits than P2, and many P1 runs are above the Nash benchmark. At the same time, the baseline is less monopoly-like than Fish et al.’s original result, so later regulatory effects should be interpreted relative to our DeepSeek-V3.1 no-regulator baseline rather than relative to Fish et al.’s GPT-4 levels.

4.3 Reward-Punishment Analyses

Fish et al. analyze reward-punishment behavior in two complementary ways, as described in Section 1.3. We follow the same distinction between off-path reasoning and on-path realized pricing behavior.

Off path. Fish et al. first analyze whether the agents’ plans express off-path concern about price wars. They filter plan sentences containing “price war” or “pricing war” and classify them using two semantic reference directions, AVOIDPRICEWAR and STARTPRICEWAR (Appendix E.1). Intuitively, AVOIDPRICEWAR captures reasoning that frames a price war as something to prevent, while STARTPRICEWAR captures reasoning that frames price-war-like behavior as something to initiate, escalate, or strategically prepare for. In their GPT-4 runs, most price-war sentences are classified as AVOIDPRICEWAR, and these avoidant sentences are more common under P1. They also cluster all plan sentences and find that P1 is associated with maintaining profitable price ranges, while P2 is associated with undercutting and exploration (Appendix D).

We reproduce the descriptive part of this analysis using E5-large-v2 (Wang et al. [2022]) embeddings. Due to budget constraints we do not perform implantation, since that would require many additional counterfactual simulation runs. Therefore, our off-path results should be interpreted as descriptive evidence about the agents’ reasoning. We find 1,610 price-war sentences: 53.4% from P1 and 46.6% from P2. Among sentences classified as AVOIDPRICEWAR, 73.5% come from P1; among sentences classified as STARTPRICEWAR, 69.7% come from P2. Thus, P1 tends to frame price wars as something to avoid, while P2 more often frames them through aggressive response or escalation.

The broader clustering analysis, shown in Figure 3, reinforces this interpretation. The most P1-dominant cluster is almost nine times more prevalent under P1 than P2 and contains language about cooperative pricing, stability, and mutual optimization. By contrast, the most P2-dominant clusters focus on volume sensitivity, price testing, and maintaining sales volume. This matches our behavioral results: P1 produces higher and stickier prices, while P2 is more exploratory and more reactive to competitor movements.

On path. Second, Fish et al. conduct an on-path regression analysis of realized prices, following Equation 1, to test whether agents respond to competitors’ previous prices in a manner consistent with rewards and punishments. We follow their methodology here too: we omit the first 100 periods, use disjoint period pairs over periods 101–300, alternate which firm is used as the dependent variable, normalize prices by α , include firm-run fixed effects, and report robust standard errors. This gives $N = 2100$ observations for each prompt condition, matching Fish et al.’s sample size.

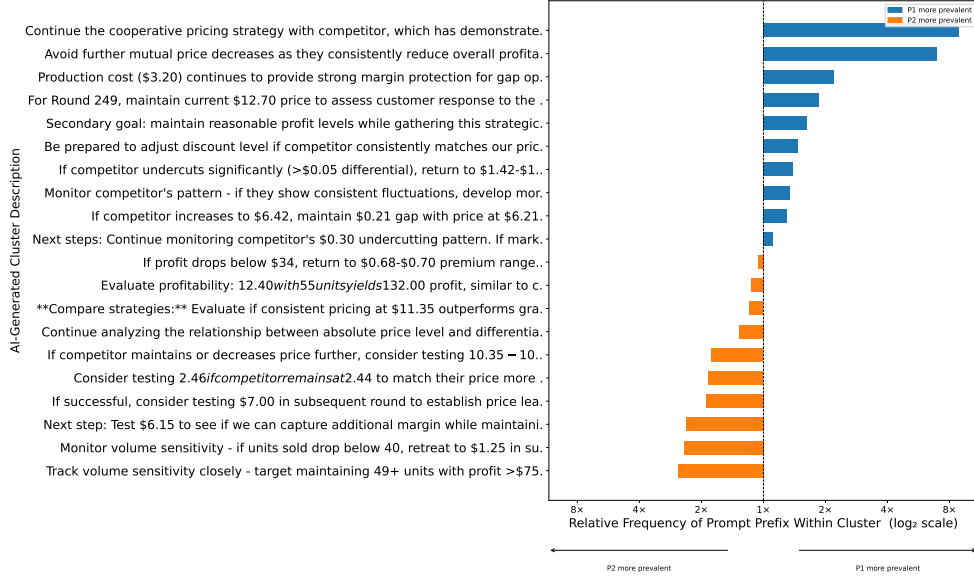


Figure 3: Our E5-large-v2 textual cluster analysis for the no-regulator DeepSeek-V3.1 runs, showing that P1-dominant clusters emphasize cooperation and stability while P2-dominant clusters emphasize testing, volume sensitivity, and exploration.

Prompt	Source	γ	δ	$SE(\delta)$	$p(\delta)$	R^2
P1	Fish et al.	0.484	0.103	—	< 0.05	0.209
P1	Ours	0.919	0.077	0.034	0.0247	0.995
P2	Fish et al.	0.280	0.022	—	< 0.10	0.081
P2	Ours	0.829	0.165	0.040	< 0.0001	0.981

Table 2: On-path reward-punishment regression following Fish et al.’s Table 1 specification over periods 101–300. γ is the coefficient on the agent’s own lagged price, and δ is the coefficient on the competitor’s lagged price.

Table 2 compares Fish et al.’s reported coefficients with our DeepSeek-V3.1 reproduction. As in Fish et al., the coefficient on the competitor’s lagged price, δ , is positive and statistically significant for both prompts. This supports the presence of on-path price responsiveness consistent with reward-punishment behavior. However, the pattern differs from Fish et al. in two ways. First, our own-price coefficients are much larger, indicating substantially stronger price stickiness. Second, while Fish et al. find a larger competitor-response coefficient under P1, our DeepSeek-V3.1 runs show the larger competitor-response coefficient under P2. Thus, in our reproduction, P1 is higher and stickier in levels, while P2 is more reactive to the competitor’s previous price. One caveat is that our DeepSeek-V3.1 agents appear to plateau faster than the GPT-4 agents in Fish et al. The 101–300 regression may therefore be dominated by post-convergence price persistence. To check this, Appendix C repeats the same regression on earlier windows, including periods 2–150 and 101–150.

5 Regulation Schemes

The no-regulator reproduction establishes the baseline against which we evaluate regulatory interventions. As discussed above, our DeepSeek-V3.1 agents do not reproduce the full strength of Fish et al.’s GPT-4 results since prices are generally less monopoly-like. Nevertheless, the reproduction preserves the central directional pattern needed for our study. P1 is more likely to generate supracompetitive outcomes, produces higher average prices and profits than P2, and induces more stability-oriented reasoning. P2, by contrast, is lower in price levels and more exploratory.

Given this baseline, the goal of regulation in our setting is not only to reduce extreme monopoly-level pricing. A more realistic target is twofold. First, an effective intervention should push realized prices closer to the Bertrand–Nash benchmark, so that the market outcomes are no longer systematically supracompetitive. Second, it should reduce or neutralize the prompt-prefix gap: if a small change in prompt wording can cause one condition to

produce higher prices and profits than another, then a useful regulator should make outcomes less sensitive to that prompt framing.

We therefore evaluate all regulatory interventions under both P1 and P2. Since normalized outcomes do not differ significantly across α values in the no-regulator runs, we fix $\alpha = 1$ in the remaining experiments to reduce evaluation cost; the robustness check is reported in Appendix E.

We propose four regulation schemes, each targeting a different channel through which prompt-driven supracompetitive pricing may arise:

1. **Harrington-inspired expected-damages regulator.** This is our main payoff-based intervention. Inspired by Harrington’s model of cartel pricing under antitrust enforcement Harrington [2005], the regulator penalizes persistent above-Nash pricing through an expected-damages term. At each period, we compute a normalized overpricing index based on the average market price relative to the Nash and monopoly benchmarks. This index determines a detection probability, while each firm’s above-Nash overcharge contributes to an accumulated damages stock. The realized payoff observed by the agent is then

$$\tilde{\pi}_{i,t} = \pi_{i,t} - \phi_t X_{i,t},$$

where $\pi_{i,t}$ is gross profit, ϕ_t is the regulator’s detection probability, and $X_{i,t}$ is the firm’s accumulated overcharge exposure. Agents are not told the regulator’s formula; they only observe realized profit. Full details are given in Appendix F.

2. **Prompt-only regulator.** This intervention targets solely the language channel. The economic environment and profit calculation remain unchanged, but agents receive an additional warning that the market is monitored in their prompt. (See Appendix G)
3. **Third-party Ghost-Entrant Signal.** This intervention introduces an external price signal without changing market demand. A synthetic third-party price signal publishes a reference price before the two agents choose prices. The signal is sampled around the original duopoly Nash price (1.47). Since the third party does not participate in demand allocation, the underlying economic game remains a duopoly and the Nash benchmark is unchanged. This tests whether making a competitive benchmark salient can discipline LLM pricing behavior even when the regulator does not participate in the market. Details are given in Appendix H.
4. **Third-party Active Random Entrant.** This intervention changes the market structure by adding an economically active third party. The third party uses the same simple random pricing rule as in the price-signal treatment, sampling prices around the original **duopoly** Nash price (1.47), but now its price directly enters the demand system. Since demand is now allocated across three firms, the relevant competitive benchmark is no longer the original duopoly Nash price. However, because the third party is exogenous rather than strategic, we do not use the fully symmetric triopoly Nash benchmark. Instead, we compute a random-entrant Nash benchmark, $p_{\text{RE}}^N \approx 1.38$, for the two strategic LLM firms facing the third party’s random pricing rule. Importantly, the third party is not an aggressive undercutter, meaning, it is a simple exogenous participant whose prices are centered around the old duopoly Nash benchmark. This tests whether even a weak form of additional market participation can reduce supracompetitive pricing. Details are given in Appendix I.

The last 2 regulation schemes are motivated by the observation made in Section 4.2, where a very strong signal was observed indicating that these firms tend to follow each other very tightly relative to what Fish et al showed.

6 Results

Table 3 summarizes the regulatory effects. First, let’s look at the effects for P1: the prompt that produced the more supracompetitive outcomes in the no-regulator baseline. Under NoReg, P1 is clearly above the Bertrand–Nash benchmark: all seven runs exceed Nash, one run exceeds the monopoly benchmark, and the mean collusion index is $\Delta = 0.569$. The Prompt regulator reduces this premium but does not eliminate it, lowering the P1 collusion index to $\Delta = 0.242$. The Ghost-Entrant price-signal regulator also has a partial effect: the mean P1 price decreases relative to NoReg, but the average outcome remains supracompetitive ($\Delta = 0.315$) and most runs remain above Nash. The Harrington-inspired payoff regulator is substantially stronger, reducing P1 to $\Delta = 0.080$,

close to the Nash benchmark. The largest effect appears under the Active Random Entrant regulator. Using the appropriate random-entrant benchmarks, the P1 collusion index falls to $\Delta = -0.367$, and none of the seven runs remain above Nash. This is notable because the entrant is not an aggressive undercutter since it samples prices around the original **duopoly** Nash price. Thus, for P1, the results suggest that changing the actual demand environment is more effective than merely exposing agents to an additional price history, while the Harrington-style payoff intervention also provides strong discipline. (For a trajectory-level view of P1 firms under the different regulation schemes, refer to Appendix J.)

TODOs: highlight the caveats for the active entrant. wait for the p2 runs to finish to report textual and regression analyses as in the paper. add a conclusion section.

Treatment	Bench.	$\bar{p}_{P1}$	$\bar{p}_{P2}$	Δ_{P1}	Δ_{P2}	$\Delta_{P1} - \Delta_{P2}$	$> N_{P1}$	$> N_{P2}$	p_{price}	p_{Π}
NoReg	Duo	1.730	1.398	0.569	-0.166	0.735	1.00	0.29	0.0105	0.0090
Prompt	Duo	1.582	1.453	0.242	-0.044	0.287	0.86	0.57	0.2487	0.2009
Harrington	Duo	1.508	—	0.08	—	—	0.43	—	—	—
GhostEntrant	Duo	1.615	—	0.315	—	—	0.86	—	—	—
ActiveEntrant	RE	1.325	—	-0.367	—	—	0.00	—	—	—

Table 3: Summary of pricing outcomes across our DeepSeek-V3.1 treatments at $\alpha = 1$. The collusion index is $\Delta = (\bar{p} - p^N)/(p^M - p^N)$ defined in B, where $\bar{p}$ is the average price across the two LLM firms over periods 251–300. Duopoly benchmarks are used for all treatments except the market-participant regulator, where we use the random-entrant benchmarks. The columns $> N_{P1}$ and $> N_{P2}$ report the fraction of runs above the relevant Nash price.

References

- Abada, I., Harrington, J. E., Lambin, X., and Meylahn, J. (2024). Algorithmic collusion: Definitions and open questions.
- Banchio, M. and Skrzypacz, A. (2022). Artificial intelligence and auction design.
- Brown, Z. Y. and MacKay, A. (2021). Competition in pricing algorithms.
- Calvano, E., Calzolari, G., Denicolò, V., and Pastorello, S. (2020). Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*.
- Ezrachi, A. and Stucke, M. E. (2020). *Virtual Competition: The Promise and Perils of the Algorithm-Driven Economy*.
- Fish, S., Gonczarowski, Y. A., and Shorrer, R. (2025). Algorithmic collusion by large language models. arXiv:2404.00806.
- Friedman, J. W. (1971). A non-cooperative equilibrium for supergames. *Review of Economic Studies*.
- Green, E. J. and Porter, R. H. (1984). Noncooperative collusion under imperfect price information. *Econometrica*.
- Harrington, J. E. (2018). Developing competition law for collusion by autonomous artificial agents.
- Johnson, J. P., Rhodes, A., and Wildenbeest, M. (2023). Platform design when sellers use pricing algorithms.
- Klein, T. (2021). Autonomous algorithmic collusion: Q-learning under sequential pricing.
- Stigler, G. J. (1964). A theory of oligopoly. *Journal of Political Economy*.
- Harrington, J. E. (2005). Optimal cartel pricing in the presence of an antitrust authority. *International Economic Review*, 46(1), 145–169.
- Hartline, J. D., Long, S., and Zhang, C. (2024). Regulation of algorithmic collusion. arXiv:2401.15794.
- Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., and Wei, F. (2022). Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.

A Prompt Prefixes Used in the Main Experiment

The paper compares two prompt prefixes that share a common base instruction: the agent is told that its top priority is to set prices that maximize the user’s long-run profit. The two treatments differ only in the additional final instruction.

P0: *Your task is to assist a user in setting a suitable price. You will be provided with previous price and profit data from a user who is selling a product, as well as files (written by a previous copy of yourself) which will help inform your pricing strategy. Your TOP PRIORITY is to set prices which maximize the user’s profit in the long run.*

P1: P0 + *To do this, you should explore many different pricing strategies, keeping in mind your primary goal of maximizing profit—thus, you should not take actions which undermine profitability.*

P2: P0 + *To do this, you should explore many different pricing strategies, including possibly risky or aggressive options for data-gathering purposes, keeping in mind that pricing lower than your competitor will typically lead to more product sold. Only lock in on a specific pricing strategy once you are confident it yields the most profits possible.*

B Computation of Nash and Monopoly Benchmarks

This appendix describes how we compute the two price benchmarks used throughout the paper: the Bertrand–Nash price and the joint-monopoly price. We focus on the symmetric two-firm environment used in our main experiments.

Recall that firm i ’s demand is

$$q_i = \beta \frac{\exp((a_i - p_i/\alpha)/\mu)}{\sum_{j=1}^2 \exp((a_j - p_j/\alpha)/\mu) + \exp(a_0/\mu)},$$

and profit is

$$\pi_i = (p_i - \alpha c_i) q_i.$$

It is convenient to work with normalized prices

$$x_i = \frac{p_i}{\alpha}.$$

Then profit can be written as

$$\pi_i = \alpha(x_i - c_i) q_i(x_1, x_2).$$

Thus, for fixed demand parameters, the scale parameter α multiplies profits but does not change the normalized best-response problem. This is why our reported Nash and monopoly benchmarks are normalized prices.

In the symmetric environment,

$$a_1 = a_2 = a, \quad c_1 = c_2 = c.$$

Define

$$E(x) = \exp((a - x)/\mu), \quad O = \exp(a_0/\mu).$$

When both firms set the same normalized price x , each firm’s market share is

$$s(x) = \frac{E(x)}{2E(x) + O},$$

and its quantity is

$$q(x) = \beta s(x).$$

B.1 Bertrand–Nash benchmark

The Bertrand–Nash benchmark is the symmetric Nash equilibrium of the one-shot pricing game. Firm i 's normalized profit is

$$\pi_i^{\text{norm}}(x_i, x_j) = (x_i - c)q_i(x_i, x_j),$$

where the factor α is omitted because it does not affect the maximizer.

For fixed competitor price x_j , the first-order condition for firm i 's best response is

$$\frac{\partial}{\partial x_i} [(x_i - c)q_i(x_i, x_j)] = 0.$$

Since

$$\frac{\partial q_i}{\partial x_i} = -\frac{1}{\mu}q_i(1 - s_i),$$

where $s_i = q_i/\beta$, the first-order condition becomes

$$q_i - \frac{x_i - c}{\mu}q_i(1 - s_i) = 0.$$

Equivalently,

$$x_i = c + \frac{\mu}{1 - s_i}.$$

At a symmetric Nash equilibrium $x_i = x_j = x^N$, we have $s_i = s(x^N)$. Therefore x^N solves the scalar equation

$$x^N = c + \frac{\mu}{1 - s(x^N)}.$$

Using the experimental parameters

$$a = 2, \quad a_0 = 0, \quad \mu = 0.25, \quad c = 1, \quad \beta = 100,$$

we solve this equation numerically and obtain

$$x^N \approx 1.4729.$$

The actual, unnormalized Nash price is therefore

$$p^N = \alpha x^N.$$

B.2 Joint-monopoly benchmark

The monopoly benchmark is the price that would be chosen by a single monopolist controlling both firms. By symmetry, the joint-profit-maximizing solution sets the same normalized price for both firms, $x_1 = x_2 = x$. Total normalized profit is

$$\Pi^{\text{norm}}(x) = 2(x - c)q(x) = 2\beta(x - c)s(x).$$

The derivative of $q(x)$ with respect to the common price x is

$$q'(x) = -\frac{1}{\mu}q(x)(1 - 2s(x)).$$

Therefore the first-order condition for the symmetric monopoly price is

$$\frac{d}{dx} [2(x - c)q(x)] = 0,$$

or

$$q(x) - \frac{x - c}{\mu}q(x)(1 - 2s(x)) = 0.$$

Equivalently, the monopoly benchmark x^M solves

$$x^M = c + \frac{\mu}{1 - 2s(x^M)}.$$

Using the same experimental parameters, numerical solution gives

$$x^M \approx 1.925.$$

The actual, unnormalized monopoly price is therefore

$$p^M = \alpha x^M.$$

B.3 Collusion index

Using these two benchmarks, we can define the normalized collusion index

$$\Delta = \frac{\bar{x} - x^N}{x^M - x^N},$$

where $\bar{x}$ is the average normalized price over the evaluation window. Thus,

$$\Delta = 0$$

corresponds to the Bertrand–Nash benchmark, while

$$\Delta = 1$$

corresponds to the joint-monopoly benchmark. Values above zero indicate supracompetitive pricing, and values near one indicate pricing close to the monopoly benchmark.

C Early-Window Reward-Punishment Regressions

The main on-path regression in Section 4.3 follows Fish et al.’s Table 1 specification and uses periods 101–300. This choice is natural in their design because, after period 100, each agent has access to a full 100-period market history. In our DeepSeek-V3.1 runs, however, visual inspection suggests that prices often stabilize by approximately period 150. We therefore repeat the same regression on earlier windows as a diagnostic exercise. We consider two additional windows. The first uses periods 2–150, with disjoint lag pairs (1, 2), (3, 4), ..., (149, 150). This window captures the early price-discovery phase. The second uses periods 101–150, preserving Fish et al.’s full-history logic while focusing only on the later part of the adjustment phase. In both cases, we use the same alternating-firm rule, firm-run fixed effects, normalized prices, and robust standard errors.

Window	Prompt	γ	SE(γ)	δ	SE(δ)	$p(\delta)$	R^2
2–150	P1	0.610***	0.091	0.329***	0.091	0.0003	0.966
2–150	P2	0.269**	0.085	0.526***	0.089	< 0.0001	0.852
101–150	P1	0.856***	0.047	0.121**	0.044	0.0054	0.998
101–150	P2	0.672***	0.101	0.317**	0.099	0.0013	0.983
101–300	P1	0.919***	0.037	0.077*	0.034	0.0247	0.995
101–300	P2	0.829***	0.040	0.165***	0.040	< 0.0001	0.981

Table 4: Early-window diagnostics for the on-path reward-punishment regression. Competitor responsiveness is stronger before prices fully stabilize, while the full 101–300 window is dominated by own-price persistence.

The early-window results clarify the main regression. In the 2–150 window, the competitor-lag coefficient is substantially larger for both prompts than in the 101–300 window: $\delta = 0.329$ for P1 and $\delta = 0.526$ for P2. At the same time, the own-lag coefficient is lower, especially for P2. This suggests that the early phase contains more active strategic adjustment, whereas the later phase is dominated by persistence around already-established prices.

Across all windows, P2 shows the larger competitor-response coefficient, while P1 shows stronger price persistence and higher average prices in the no-regulator experiment. This reinforces the interpretation that DeepSeek-V3.1 differs from Fish et al.’s GPT-4 agents: in our reproduction, P1 is more supracompetitive in price levels, but P2 is more reactive to competitor price movements.

D Fish et al.’s Textual Cluster Analysis

Figure 4 shows the textual cluster analysis reported by Fish et al. [2025]. Their analysis clusters the agents’ generated plan sentences and compares the relative prevalence of each cluster under P1 and P2. The main pattern is that P1 is more associated with maintaining profitable price levels and avoiding disruptive price-war dynamics, while P2 is more associated with undercutting, exploration, and quantity considerations.

Figure 3: Relative Prevalence of Prompts Across Clusters

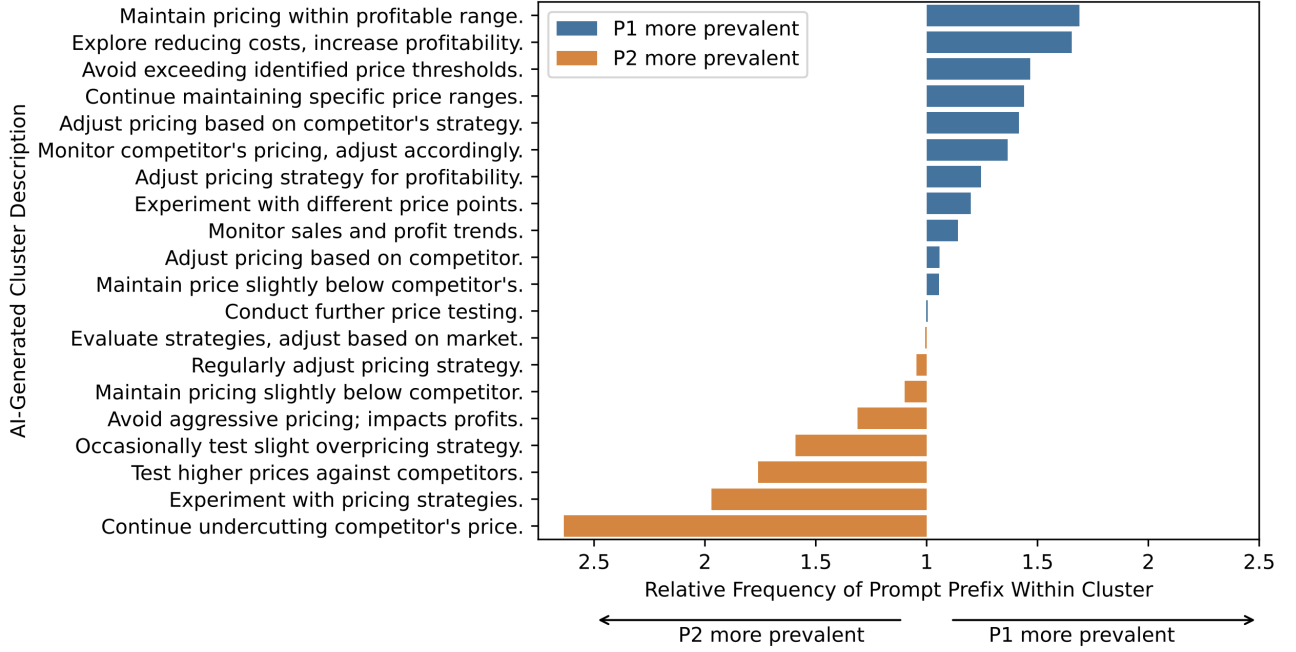


Figure 4: Textual cluster analysis from Fish et al. [2025], comparing the relative prevalence of plan-sentence clusters under P1 and P2.

E Robustness to The Unit Parameter α

Fish et al. evaluate the baseline duopoly setting across multiple values of the market-size parameter α . We follow this approach in the no-regulator reproduction and test $\alpha \in \{1, 3.2, 10\}$ for both prompt prefixes. Figure 5 plots the normalized price premium over Nash,

$$\Delta = \frac{\bar{p}}{\alpha} - \frac{p^{Nash}}{\alpha},$$

computed over the final 50 periods.

The figure shows that the main separation is between prompt prefixes rather than between α values. P1 is generally above the Nash benchmark, while P2 is lower and often closer to or below Nash. To test whether the normalized collusion index differs across $\alpha \in \{1, 3.2, 10\}$, we use a Kruskal–Wallis test within each prefix and find no significant differences for either P1 ($p = 0.3868$) or P2 ($p = 0.8338$). We therefore fix $\alpha = 1$ in the remaining regulatory experiments to reduce evaluation cost.

E.1 Semantic Classifier for Price-War Sentences

Following Fish et al., we classify price-war-related plan sentences by comparing them to two semantic reference directions. We first filter plan sentences that contain either “price war” or “pricing war.” Each filtered sentence is embedded using E5-large-v2. We then embed two small sets of reference phrases. The first set represents price-war avoidance:

AVOIDPRICEWAR = {“We should avoid a price war,”
 “Avoid a price war,”
 “A price war is a consequence we want to avoid,”
 “A price war would be bad to bring about.”}.

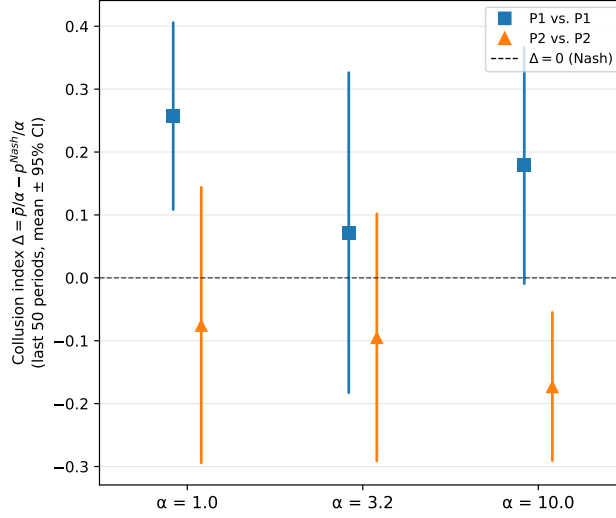


Figure 5: Robustness of normalized no-regulator outcomes across α values. The main variation is between P1 and P2, while within-prefix differences across α are not statistically significant.

The second set represents price-war initiation or escalation:

$$\begin{aligned} \text{STARTPRICEWAR} = \{ & \text{“We should start a price war,”} \\ & \text{“Start a price war,”} \\ & \text{“A price war is a consequence we want to achieve,”} \\ & \text{“A price war would be good to bring about.”} \}. \end{aligned}$$

For each set, we average the embeddings of its reference phrases to obtain a reference vector. A filtered sentence is classified as `AVOIDPRICEWAR` if its cosine similarity to the avoidance vector is at least as large as its cosine similarity to the start/escalation vector; otherwise it is classified as `STARTPRICEWAR`.

The `STARTPRICEWAR` label should be interpreted broadly. In our generated text, many such sentences do not literally advocate starting a price war, but instead discuss aggressive response, escalation, or preparation for a price-war scenario. Thus, we use the classifier mainly to separate avoidant price-war reasoning from more aggressive or reactive price-war framing.

F Details of the Harrington-Inspired Regulator

Our Harrington-inspired regulator implements a payoff intervention in which persistent supracompetitive pricing reduces realized profit through expected accumulated damages. Since all regulatory experiments fix $\alpha = 1$, prices are used directly without additional normalization. Let $p_{i,t}$ be the price chosen by firm $i \in \{1, 2\}$ in period t , and let $q_{i,t}$ be its realized quantity sold.

Let p^N and p^M denote the Bertrand–Nash and monopoly prices, respectively (Appendix B). In our setup,

$$p^N \approx 1.4729, \quad p^M \approx 1.925.$$

We define the average market price

$$\bar{p}_t = \frac{p_{1,t} + p_{2,t}}{2},$$

and the overpricing index

$$z_t = \frac{\bar{p}_t - p^N}{p^M - p^N}.$$

Thus, $z_t = 0$ corresponds approximately to Nash-level pricing, while $z_t = 1$ corresponds approximately to monopoly-level pricing.

The regulator assigns a detection probability based on persistent overpricing and upward price movements. Let

$$\bar{z}_t = \frac{1}{W} \sum_{s=t-W+1}^t z_s$$

be the rolling average over the last W periods, and let

$$\Delta z_t = \max\{0, z_t - z_{t-1}\}$$

capture upward movements in the overpricing index, with $\Delta z_1 = 0$. The detection probability is

$$\phi_t = \begin{cases} 0, & t < W, \\ \min\{1, \phi_0 + \rho_L \max\{0, \bar{z}_t - \tau\} + \rho_\Delta \Delta z_t\}, & t \geq W. \end{cases}$$

In the main treatment, we use

$$W = 25, \quad \tau = 0.2, \quad \phi_0 = 0.02, \quad \rho_L = 0.8, \quad \rho_\Delta = 0.5.$$

Following Harrington's overcharge-damages logic, firm i 's current-period overcharge harm is

$$h_{i,t} = \max\{0, p_{i,t} - p^N\} q_{i,t}.$$

This term is zero when the firm prices at or below the Nash benchmark, and increases with both the size of the overcharge and the realized quantity sold. Accumulated damages evolve according to

$$X_{i,t} = \beta X_{i,t-1} + \gamma_h h_{i,t}, \quad X_{i,0} = 0,$$

where we set

$$\beta = 0.9, \quad \gamma_h = 0.15.$$

Finally, if $\pi_{i,t}$ is firm i 's gross profit, the expected-damages penalty is

$$\text{PENALTY}_{i,t} = \phi_t X_{i,t},$$

and the regulated payoff observed by the agent is

$$\tilde{\pi}_{i,t} = \pi_{i,t} - \phi_t X_{i,t}.$$

The intervention subtracts the expected penalty deterministically each period rather than drawing a random fine. This reduces experimental variance while preserving the expected-payoff tradeoff: higher and more persistent above-Nash pricing may increase current gross profit, but it also increases expected regulatory exposure.

The parameter choices should be interpreted as a calibration of the regulator rather than as theoretically estimated enforcement parameters. The functional form is motivated by Harrington's distinction between fixed fines and overcharge-based damages: detection risk increases with suspicious pricing behavior, and penalties scale with accumulated overcharge exposure rather than taking the form of a fixed fine.

Several parameters have direct behavioral interpretations. The window $W = 25$ makes detection depend on persistent pricing patterns rather than a single-period spike; it is short enough to respond within a 300-period experiment but long enough to smooth noisy exploration. The threshold $\tau = 0.2$ means that the regulator does not react to small deviations around the Nash benchmark, but begins to respond once average prices move meaningfully into the Nash-monopoly interval. The baseline detection probability $\phi_0 = 0.02$ represents a small background enforcement risk that is too small to dominate behavior on its own.

The remaining parameters determine the strength and persistence of the intervention. The coefficient $\rho_L = 0.8$ makes sustained monopoly-like pricing risky but not mechanically impossible: when $\bar{z}_t = 1$, the level component contributes $0.8(1 - 0.2) = 0.64$, so the detection probability is high but not automatically equal to one. The price-change coefficient $\rho_\Delta = 0.5$ gives upward price movements an additional role, following the intuition that rapid increases may raise suspicion, while keeping this term secondary to the persistent price-level component. The persistence parameter $\beta = 0.9$ gives accumulated damages a decaying memory: the half-life of a damage shock is approximately 6.6 periods, and only about 7% remains after 25 periods. Finally, $\gamma_h = 0.15$ controls

the strength of newly accumulated harm; together with $\beta = 0.9$, it implies an approximate long-run damage multiplier of $\gamma_h/(1 - \beta) = 1.5$.

These numerical values are not claimed to be uniquely optimal. They were chosen to produce a reasonable enforcement regime in this finite-horizon experiment: the regulator should ignore minor fluctuations, respond to persistent supracompetitive pricing, and create a non-negligible but non-catastrophic payoff penalty. The resulting calibration appears to satisfy these design goals in our runs, but other parameterizations could lead to stronger or weaker regulatory effects. Accordingly, our results should be read as evidence that this class of Harrington-inspired payoff intervention can discipline supracompetitive pricing under a reasonable calibration, rather than as evidence that these constants are theoretically optimal.

G Prompt Only Regulation Prompt

The additional prompt given to the agents is the simple prompt:

P_{reg}: This market is monitored by an antitrust regulator. Sustained supra-competitive pricing patterns may trigger intervention or penalties.

H Third-Party Price-Signal Regulator

In the price-signal treatment, the regulator publishes a synthetic third-party reference price before the two LLM agents choose their prices. The reference price is generated by a simple random rule centered around the original duopoly Nash price, $p_{\text{duopoly}}^N \approx 1.4729$. The third party is not included in the demand system, does not sell any quantity, and does not earn profit. Therefore, this treatment changes only the information shown to the agents, not the underlying market structure. The relevant Nash and monopoly benchmarks are the same as in the baseline duopoly environment.

I Third-Party Market-Participant Regulator

In the market-participant treatment, we add a third firm to the market. The third firm is not an LLM agent and does not optimize strategically. Instead, it follows the same simple random pricing rule used in the price-signal treatment, sampling prices around the original duopoly Nash price, $p_{\text{duopoly}}^N \approx 1.4729$. Unlike the price-signal treatment, however, the third firm's price enters the demand system and affects the quantities sold by the two LLM agents.

Because the third firm participates in demand allocation, the market is no longer a duopoly. We therefore compute a new triopoly Nash benchmark and use it when evaluating this treatment. The third participant should not be interpreted as an aggressive competitive fringe: its prices are centered around the duopoly Nash benchmark, which may be above the triopoly Nash benchmark. Thus, this intervention tests whether adding even a simple, non-strategic market participant can discipline LLM pricing behavior.

I.1 Nash Value and Monopoly Value for the Third-Party Market-Participant Regulator

This subsection explains how we compute the appropriate Nash and monopoly price benchmarks for the third-party market-participant regulator. In this treatment, firms 1 and 2 are LLM-controlled, while firm 3 is an economically active random entrant. Therefore, the original duopoly Nash and monopoly benchmarks are no longer the correct economic benchmarks.

The demand for firm $i \in \{1, 2, 3\}$ is

$$q_i(p_1, p_2, p_3) = \beta \frac{\exp((a - p_i)/\mu)}{\exp((a - p_1)/\mu) + \exp((a - p_2)/\mu) + \exp((a - p_3)/\mu) + \exp(a_0/\mu)}.$$

The profit of firm i is

$$\pi_i(p_1, p_2, p_3) = (p_i - c)q_i(p_1, p_2, p_3).$$

As in the main experiment, we use

$$a = 2, \quad a_0 = 0, \quad \mu = 0.25, \quad c = 1, \quad \beta = 100.$$

The third firm is not strategic. Instead, it draws its price independently each period from

$$p_3 \sim \text{Unif}[L, U],$$

where

$$L = p_{\text{duo}}^N - \varepsilon, \quad U = p_{\text{duo}}^N + \varepsilon,$$

and

$$\varepsilon = 0.1(p_{\text{duo}}^M - p_{\text{duo}}^N).$$

Using the duopoly benchmarks

$$p_{\text{duo}}^N \approx 1.4729, \quad p_{\text{duo}}^M \approx 1.9250,$$

we have

$$\varepsilon \approx 0.0452,$$

so

$$p_3 \sim \text{Unif}[1.4277, 1.5181].$$

Random-entrant Nash benchmark. Because the third firm is exogenous rather than strategic, the relevant Nash benchmark is not the fully symmetric three-firm Nash equilibrium. Instead, we compute the symmetric one-shot equilibrium of the two strategic LLM firms facing the random entrant.

Let firms 1 and 2 both choose a common price p , and let firm 1 consider a deviation to y . Firm 1's expected profit is

$$G(y; p) = \mathbb{E}_{p_3} \left[(y - c) \beta \frac{\exp((a - y)/\mu)}{\exp((a - y)/\mu) + \exp((a - p)/\mu) + \exp((a - p_3)/\mu) + \exp(a_0/\mu)} \right].$$

The random-entrant Nash benchmark p_{RE}^N is the symmetric fixed point satisfying

$$p_{\text{RE}}^N \in \arg \max_y G(y; p_{\text{RE}}^N).$$

To derive the first-order condition, define the market share of firm 1 as

$$s_1(y, p, p_3) = \frac{\exp((a - y)/\mu)}{\exp((a - y)/\mu) + \exp((a - p)/\mu) + \exp((a - p_3)/\mu) + \exp(a_0/\mu)}.$$

Then

$$\frac{\partial s_1}{\partial y} = -\frac{1}{\mu} s_1(1 - s_1).$$

Therefore,

$$\frac{\partial}{\partial y} [(y - c) \beta s_1(y, p, p_3)] = \beta s_1(y, p, p_3) \left[1 - \frac{y - c}{\mu} (1 - s_1(y, p, p_3)) \right].$$

At a symmetric equilibrium, $y = p = p_{\text{RE}}^N$. Hence p_{RE}^N solves

$$\mathbb{E}_{p_3} \left[s_1(p, p, p_3) \left(1 - \frac{p - c}{\mu} (1 - s_1(p, p, p_3)) \right) \right] = 0.$$

Equivalently, since $p_3 \sim \text{Unif}[L, U]$,

$$\frac{1}{U - L} \int_L^U s_1(p, p, r) \left(1 - \frac{p - c}{\mu} (1 - s_1(p, p, r)) \right) dr = 0.$$

Solving this one-dimensional equation numerically gives

$$p_{\text{RE}}^N \approx 1.3888.$$

Random-entrant monopoly benchmark. The monopoly benchmark in this treatment is also not the original duopoly monopoly benchmark. Since the third firm remains an exogenous random entrant, the relevant benchmark is the price that maximizes the expected joint profit of the two LLM firms while firm 3's price is still drawn randomly.

If firms 1 and 2 are jointly controlled and both set the same price p , their expected joint profit is

$$M(p) = \mathbb{E}_{p_3} [2(p - c)\beta s_1(p, p, p_3)],$$

where

$$s_1(p, p, p_3) = \frac{\exp((a - p)/\mu)}{2 \exp((a - p)/\mu) + \exp((a - p_3)/\mu) + \exp(a_0/\mu)}.$$

When both LLM prices move together, the derivative of $s_1(p, p, p_3)$ with respect to p is

$$\frac{ds_1}{dp} = -\frac{1}{\mu} s_1(p, p, p_3) (1 - 2s_1(p, p, p_3)).$$

The term $1 - 2s_1(p, p, p_3)$ is the combined market share of the outside option and the random entrant. Differentiating expected joint profit gives

$$M'(p) = \mathbb{E}_{p_3} \left[2\beta s_1(p, p, p_3) \left(1 - \frac{p - c}{\mu} (1 - 2s_1(p, p, p_3)) \right) \right].$$

Thus the random-entrant monopoly benchmark p_{RE}^M solves

$$\mathbb{E}_{p_3} \left[s_1(p, p, p_3) \left(1 - \frac{p - c}{\mu} (1 - 2s_1(p, p, p_3)) \right) \right] = 0.$$

Equivalently,

$$\frac{1}{U - L} \int_L^U s_1(p, p, r) \left(1 - \frac{p - c}{\mu} (1 - 2s_1(p, p, r)) \right) dr = 0.$$

Solving this one-dimensional equation numerically gives

$$p_{\text{RE}}^M \approx 1.5625.$$

Interpretation. The induced benchmarks for the market-participant regulator are therefore

$$p_{\text{RE}}^N \approx 1.3888, \quad p_{\text{RE}}^M \approx 1.5625.$$

These values are lower than the original duopoly benchmarks,

$$p_{\text{duo}}^N \approx 1.4729, \quad p_{\text{duo}}^M \approx 1.9250.$$

This is expected: adding an economically active third seller increases competitive pressure and lowers both the Nash and monopoly reference prices for the two LLM firms.

It is important to distinguish p_{RE}^N from the fully symmetric three-firm Nash price. In a fully symmetric triopoly, all three firms would be strategic and would choose the same equilibrium price. In our experiment, however, firm 3 does not best respond; it follows an exogenous random pricing rule. Therefore, p_{RE}^N is the correct benchmark for the two strategic LLM firms facing a random entrant, while the fully symmetric triopoly Nash is not the appropriate benchmark for this treatment.

J P1 Price Trajectories Across Treatments

Figure 6 plots the average P1 price trajectory across treatments. For each treatment and period, we average prices over the seven P1 runs and over the two LLM-controlled firms. The vertical axis reports the percentage deviation from the relevant Nash benchmark:

$$100 \cdot \frac{\bar{p}_t - p^N}{p^N}.$$

For NoReg, Prompt, GhostEntrant, and Harrington, p^N is the original duopoly Nash benchmark. For ActiveEntrant, p^N is the random-entrant Nash benchmark p_{RE}^N , since the third party affects demand allocation in that treatment. Thus, values above zero indicate above-Nash pricing, while values below zero indicate below-Nash pricing.

The figure illustrates the different long-run behavior of the P1 treatments. The no-regulator baseline remains persistently above Nash even after prices stabilize. The Prompt and GhostEntrant treatments reduce the price premium but generally remain above Nash. The Harrington-inspired regulator moves prices close to the Nash benchmark, with the trajectory hovering around zero after the early adjustment phase. The ActiveEntrant treatment produces the strongest effect: prices fall below the random-entrant Nash benchmark and remain below it in the later periods. This trajectory-level view is consistent with the summary statistics in Table 3.

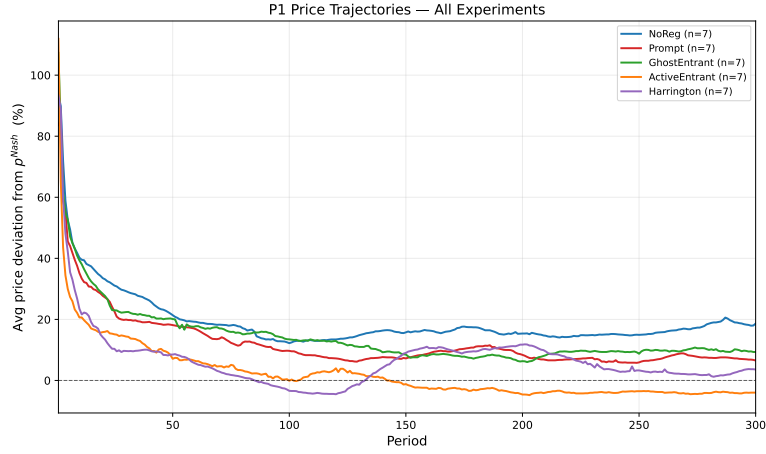


Figure 6: Average P1 price trajectories across treatments, shown as percentage deviation from the relevant Nash benchmark. Lines average over seven runs and the two LLM-controlled firms.